LEONARDO **VICENTINI**

Software Engineer

Open to relocation across the EU, UK & CH

(+39) 3453168243 | vicentini.leonardo99@gmail.com | leonardovicentini.com | GitHub: vicentinileonardo | LinkedIn: leonardovicentini

Skills

Programming Go, Python, Java, JavaScript, Bash, SQL
Platform & Cloud Kubernetes, Kubernetes Operators, OpenAPI, REST APIs, Docker, Helm, AWS, GCP, Azure, Linux
DevOps & MLOps GitOps, Ansible, CI/CD, Prometheus, Grafana, MLflow, KServe

Education

University of Trento
Trento, Italy

Master's degree in Computer Science (English) — Grade: 110/110 with honors (GPA: 29.73/30) *Sep. 2021 — Mar. 2025*

- Relevant courses: Distributed Systems, Cloud Computing, Service Design, Security Testing, Blockchain, Data Mining, HPC

Bachelor's degree in Computer Science — Grade: 106/110 *Sep. 2018 — Sep. 2021*

- Relevant courses: Algorithms & Data Structures, Software Engineering, OOP, Operating Systems, Databases, Networks, HCI, ML

Work Experience

Krateo - Cloud platform startup supporting enterprise clients
Verona, Italy (Remote)

Software Engineer - Kubernetes, Cloud & Platform *May 2025 — Jun. 2026*

- Owned the end-to-end development of core components of the Kubernetes-based Krateo platform, from design through production.
- Built a Postgres-backed query API in Go with RBAC-aware filtering and keyset pagination, eliminating slow on-demand resource queries that degraded frontend UX, cutting latency from ∼7s to ∼400ms and sustaining 270+ req/s, validated via 150+ load-test runs.
- Evolved a Kubernetes Operator framework that auto-generates CRDs and controllers from OpenAPI specs, eliminating hand-written operator code for APIs not natively managed in Kubernetes and isolating failures per resource type.
- Independently migrated and extended 2 legacy integrations (Azure DevOps, GitHub) onto the OpenAPI-driven operator framework, growing coverage to 20 production resource types published to the platform marketplace.
- Provided rotational L1–L3 support for 4 enterprise deployments spanning 12 Kubernetes clusters (dev/test/prod), diagnosing incidents and performing root cause analysis across clusters, platform components, and external integrations.
- Designed, documented, and oversaw rollout of near-zero-downtime platform upgrades across 4 enterprise deployments, validated against replicated client environments and codified into reusable runbooks.
- Evaluated frameworks (kagent, Google ADK), memory systems, and MCP/A2A protocols for Krateo Autopilot, a multi-agent system.

Master thesis researcher - Kubernetes, Cloud & Platform — [**Preprint**] *Oct. 2024 — Apr. 2025*

- Integrated Open Policy Agent as a Kubernetes mutating webhook to enforce latency-SLA, GDPR data-residency, and carbon-aware scheduling policies, contributing to a system achieving up to 13% CO2 emission reductions vs. round-robin baseline scheduling.
- Developed Helm templates that dynamically select the best-fit VM size, gate provisioning on cloud provider and scheduling time, and drive VM and networking infrastructure creation across AWS, GCP, and Azure via native Kubernetes Operators.
- Architected and deployed an MLOps pipeline serving ∼15 PyTorch carbon-forecasting models using MLflow and KServe on K8s.

ESA – European Space Agency (European Space Operations Centre)
Darmstadt, Germany

Software Engineer Intern - Cloud & Infrastructure *Apr. 2024 — Jul. 2024*

- Delivered a GitOps-style Service Status Board with a GitLab CI/CD pipeline enforcing schema validation on incident, intervention, and announcement metadata before publishing, used by 3 Space Mission Operations teams to track critical infrastructure systems.
- Designed a Prometheus-based instrumentation layer with self-registering targets, spanning 5 exporter types (OS, Windows, process, network/SNMP, and uptime checks) across 10+ infrastructure sources and 4 Linux distributions.
- Built 15 Grafana dashboards visualizing time-series infrastructure metrics and an Alertmanager notification pipeline via Telegram, delivering near real-time health visibility for space mission operations.
- Automated secure deployment and TLS provisioning for monitoring agents via Ansible across 10s of VMs in restricted Operations LAN.
- Deployed 3 production instances of the monitoring platform using Kubernetes and Helm following a cloud-native architecture.

FIPIC – Italian Wheelchair Basketball Federation
Rome, Italy (Remote)

Software Engineer Intern - Backend *Feb. 2021 — Jun. 2021*

- Led a team of 4 engineers to develop a federation-wide historical data and multimedia archive, reducing delivery time by ∼50% by improving planning, task distribution, and development workflows.
- Built an ELK-based data pipeline powering 4 dashboards, and exposed 70+ Node.js REST endpoints over a MySQL database.

Projects

Multi-level distributed cache

Java, Akka — [**Code** | **Report**]

- Designed a 2-tier cache hierarchy (client, L2, L1, database) in Akka with 4 message-passing consistency protocols, including a custom 4-phase propose/accept/apply/confirm write protocol guaranteeing no cache observes a stale value system-wide.
- Built a bandwidth-aware crash-recovery algorithm that reconciles only stale keys via targeted re-fetch and comparison, plus an HTTP-driven tool that verifies system-wide consistency by cross-checking every cache's state against the database.